Hydrofarms

Agri Solutions

NFT Nutrient Feed and Return Network

Contractor Installation Guide and Detailed Quantity Takeoff

Main greenhouse	Operating circuits	Document status
126 x 36 m	East + West + Nursery	Construction coordination and procurement issue

Routes and quantities are based on the latest approved CAD measurements recorded through 11 August 2026.

1. Purpose and Scope Boundaries

This guide describes the complete nutrient-solution circuit from each operating tank to all production and nursery NFT tables, and back to the same tank. It gives pipe routes, lengths, fittings, installation requirements, testing and handover records.

Feed network starts
At each 3-in pump-set discharge flange, downstream of pump-package accessories.

Feed network ends
At two nutrient entry points on every NFT channel: the inlet end cap and channel midpoint.

Return network starts
At each NFT drain cap, through the table collector, downpipe and buried gravity network.

Return network ends
At the dedicated operating-tank return-entry kit, including sleeve, waterstop and flexible connection.

Design basis

78 tables
Production: 10 NFT channels per table, 16 m long.

6 tables
Nursery: 17 NFT channels per table, 16 m long.

882 channels
780 production + 102 nursery.

1,764 points
Two feed points per channel.

3 independent circuits
East production, west production and nursery.

84 table interfaces
Independent feed and return connection for each table.

Mandatory operating rule: no circuit return may enter another circuit's tank. East returns only to the east tank, west only to the west tank, and nursery only to the nursery tank.

BOQ ownership and non-duplication

Pump check valves, suction strainers, base flexible connectors and pump instruments are measured once in the pump-set package. This guide measures the network interface and all downstream feed components. Civil excavation, backfilling and concrete are measured in the civil chapter; sleeves, waterstops and tank-penetration pieces remain in this piping package.

2. Complete Flow Logic

Operating tank → 4-in suction → Pump + VFD → 3-in discharge → Buried 3-in feed → 1-in table takeoff

1-in table header → Two 3/4-in manifolds → 5/16-in valves and tubes → NFT channel → 1-in drain branch → 3-in collector/downpipe

Buried 4-in return → 6-in production main → Root trap → Tank entry kit → Same operating tank

Pump groups at the network interface

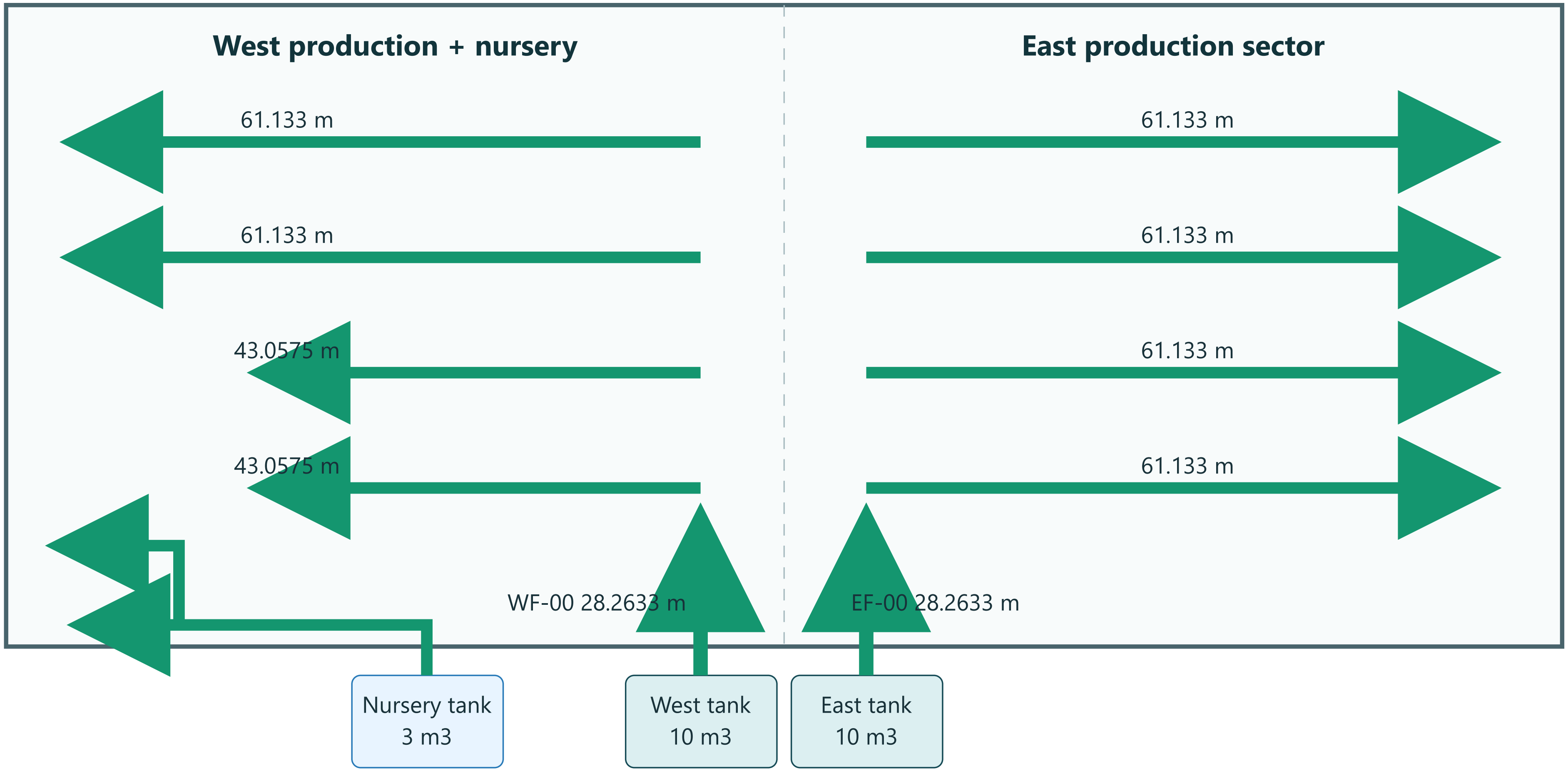
Circuit	Pumps	Duty point per pump	Suction	Discharge	Operating arrangement
East and west production	3	30 m3/h at 30 m TDH - approx. 7.5 HP / 5.5 kW	4 in	3 in	Two duty pumps, one per tank, plus one common standby through selection valves
Nursery	2	8 m3/h at 25 m TDH - approx. 2 HP / 1.5 kW	4 in	3 in	One duty + one standby

Commissioning target: normal NFT channel flow is 1.2-1.5 L/min, divided approximately equally between the two feed entries. Final balancing is by the mini valves after checking the nearest and farthest channels.

Circuit separation

The three circuits are hydraulically independent. No cross-connection is permitted except at the intentionally designed common standby-pump manifold, which remains closed and tagged under normal operation. Clear pipe-surface spacing between east and west 3-in feed mains is 2.235 m; centre-to-centre spacing is approximately 2.325 m using a nominal 90 mm outside diameter.

3. Buried Feed Network Plan



South greenhouse boundary / tank-room zone - arrows indicate pressurised feed direction

Drawing convention: green lines are pressurised 3-in feed pipes. Each circuit leaves its pump south of the greenhouse, runs north and branches east or west toward table groups. All tee chainages are centre-to-centre dimensions.

East circuit
Four 61.133 m branches at main chainages 1.7000, 9.7633, 20.2783 and 28.2633 m.

West circuit
Four branches at the same chainages: the first two are 43.0575 m and the third and fourth are 61.133 m.

4. Feed Route Schedule

Code	Route	Diameter	Direction from pump	Length m	Setting-out note
EF-00	East production - pump discharge main entering the greenhouse	3 in	North	28.2633	Branch tee chainages: 1.7000, 9.7633, 20.2783 and 28.2633 m
EF-01	East production - branch 1	3 in	East	61.1330	From tee 1 to the end of the table group
EF-02	East production - branch 2	3 in	East	61.1330	From tee 2 to the end of the table group
EF-03	East production - branch 3	3 in	East	61.1330	From tee 3 to the end of the table group
EF-04	East production - branch 4	3 in	East	61.1330	From tee 4 to the end of the table group
WF-00	West production - pump discharge main entering the greenhouse	3 in	North	28.2633	Same branch chainages as the east circuit
WF-01	West production - branch 1	3 in	West	43.0575	To the end of the table group
WF-02	West production - branch 2	3 in	West	43.0575	To the end of the table group
WF-03	West production - branch 3	3 in	West	61.1330	To the end of the table group
WF-04	West production - branch 4	3 in	West	61.1330	To the end of the table group
NF-01	Nursery - pump discharge into greenhouse	3 in	North	1.4400	Then one 90-degree elbow west
NF-02	Nursery - west main / first branch	3 in	West	55.3059	Second-branch tee at 39.248 m from elbow centre
NF-03	Nursery - second-branch stem	3 in	North	8.0370	From tee centre, then elbow west
NF-04	Nursery - second-branch terminal leg	3 in	West	16.0899	Removable flushing end

Setting-out and installation instructions

1. Use the centre of each pump discharge flange as the route datum and transfer pipe centrelines to site before excavation.
2. Fix permanent marks at every tee, elbow and terminal, and coordinate them with table legs, aisles and foundations before concrete placement.
3. Use uPVC pressure PN16 with socket solvent-weld fittings from one compatible system wherever practical.
4. Use long elbows or properly swept routes; do not fabricate sharp elbows from short cut pieces.
5. Provide concrete thrust restraint at tees and bends where required by the pipe manufacturer or pressure test, while keeping joints visible until acceptance.

Contractor alert: the adopted production-main length is 28.2633 m, not an approximate 28 m. The nursery tee is 39.248 m from the centre of the first west-turn elbow.

5. On-Table Feed Assembly

Component sequence

1. 3 x 1-in reducing tee or approved saddle from buried feed main.
2. Accessible valve box containing one 1-in full-bore ball valve and one 1-in union.
3. One 1-in riser independently clamped to the table chassis.
4. One 1-in longitudinal header supplying two 3/4-in transverse manifolds: one at the channel inlet and one at the 8.00 m midpoint.
5. Each manifold includes a 3/4-in isolation valve, union and removable flushing end.
6. Each NFT entry receives one 5/16-in tube through a mini valve, barbed entry fitting and EPDM grommet.

One production table
10 channels x 2 entries = 20 tubes, 20 mini valves and 20 entry fittings.

One nursery table
17 channels x 2 entries = 34 tubes, 34 mini valves and 34 entry fittings.

Valve elevation
Locate manifold valves approximately 150 mm below the NFT channels, with clear hand access.

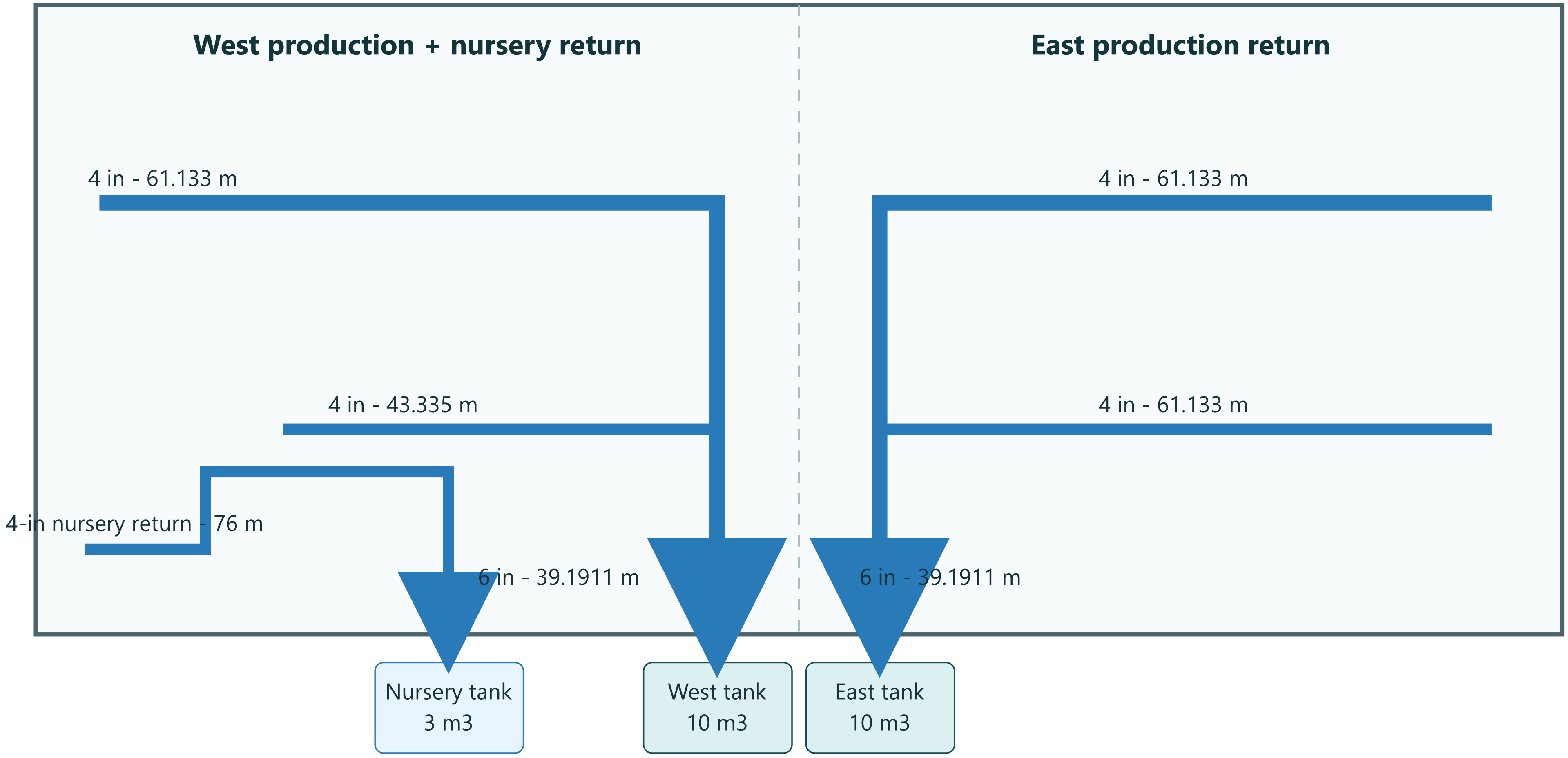
Area	Tables	Channels/table	Total channels	Points/channel	Total feed points
Production tables	78	10	780	2	1,560
Nursery tables	6	17	102	2	204
Total	84	-	882	-	1,764

On-table feed quantity takeoff

Item	Unit	Installed	Purchase	Quantity basis
1-in uPVC PN16 feed riser	pc	84	93	From isolation box to table header
1-in uPVC PN16 longitudinal/side header	m	1,344	1,482	16 m per table; 247 commercial pipes x 6 m
3/4-in uPVC PN16 transverse manifold	m	336	372	2 manifolds x 2 m per table; 62 pipes x 6 m
3/4-in full-bore ball valve	pc	168	185	One per manifold
3/4-in union	pc	168	185	Manifold removal
3/4-in removable flushing end	pc	168	185	Manifold flushing
5/16-in PE microtube	m	1,764	1,941	Supply 1 m per feed point and trim on site
5/16-in barbed mini ball valve	pc	1,764	1,941	Individual feed-point balancing
5/16-in barbed entry fitting + EPDM grommet	set	1,764	1,941	Watertight NFT-channel entry
UV-resistant / rubber-lined stainless pipe clamps	table set	84	93	Independent support from NFT channels

Balancing acceptance: after flushing, adjust the mini valves so the measured flow difference between representative nearest and farthest feed points remains within +/-10% of target.

6. Gravity Return Network Plan



Arrows show gravity flow from tables to operating tanks - the three circuits remain independent

Production return

Each 3-in table downpipe joins a 4-in branch through a 45-degree wye. Two 4-in collection branches join one independent 6-in return main for each production tank.

Nursery return

A fully independent 4-in route, 76 m horizontal measured length, runs from the remote cleanout to the nursery tank without joining production return lines.

No return pump: this is a gravity system. Performance depends on continuous fall, no backfall, 45-degree entries and partially full flow.

7. Return Route Schedule

Code	Route	Diameter	Reading/flow direction	Length m	Note
ER-M	East production return main to east tank	6 in	South to tank	39.1911	Two 4-in branches enter at 20.456 and 39.000 m from tank-entry centre
ER-01	East production return branch 1	4 in	Toward ER-M	61.1330	Connect through 45-degree reducing wye
ER-02	East production return branch 2	4 in	Toward ER-M	61.1330	Connect through 45-degree reducing wye
WR-M	West production return main to west tank	6 in	South to tank	39.1911	Same collection chainages as east circuit
WR-01	West production return branch 1	4 in	Toward WR-M	43.3350	Connect through 45-degree reducing wye
WR-02	West production return branch 2	4 in	Toward WR-M	61.1330	Connect through 45-degree reducing wye
NR-01	Nursery - tank to first elbow, measured opposite to flow	4 in	North	4.3000	Actual flow is in reverse direction toward tank
NR-02	Nursery - second leg	4 in	West	37.9000	Long-radius elbow at end
NR-03	Nursery - third leg	4 in	North	16.9000	Long-radius elbow at end
NR-04	Nursery - remote terminal leg	4 in	West	16.9000	Cleanout at remote end

Slopes and levels

3-in table collector

1.0%

Toward downpipe.

Buried 4-in and 6-in return

0.7%

Design slope; never less than 0.5%.

Survey check

Every 10 m

Record invert level.

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Slope (%) = (Upstream invert - Downstream invert) / Horizontal length x 100
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Cleanouts and tank entry

- Provide access at each remote end, major direction change and at a maximum guidance spacing of 20-25 m.
- Install a root/debris trap with a lift-out stainless basket and 1/8-in openings before each operating tank.
- Return entry shall include a downward internal elbow to limit splashing and promote stable tank circulation.
- Use a flexible connector and dismantling joint before tank penetration to isolate settlement and pipe stress.

8. Final Pipe-Length Takeoff

System	Material/class	Installed length m	Purchase quantity	Basis
East production feed	3-in uPVC pressure PN16	272.7953	Included in 3-in total	CAD centreline measurement
West production feed	3-in uPVC pressure PN16	236.6443	Included in 3-in total	CAD centreline measurement
Nursery feed	3-in uPVC pressure PN16	80.8728	Included in 3-in total	CAD centreline measurement
Risers and tank-room entry allowance	3-in uPVC pressure PN16	3.0000	-	1 m for each independent circuit
3-in installed total	3-in uPVC pressure PN16	593.3124	654 m = 109 pipes x 6 m	10% allowance, rounded to full commercial lengths
East production return branches	4-in heavy-duty uPVC drainage SN8	122.2660	Included in 4-in total	2 x 61.133 m
West production return branches	4-in heavy-duty uPVC drainage SN8	104.4680	Included in 4-in total	43.335 + 61.133 m
Nursery return	4-in heavy-duty uPVC drainage SN8	76.0000	Included in 4-in total	Independent circuit
Nursery tank-room entry allowance	4-in heavy-duty uPVC drainage SN8	1.0000	-	Tank entry/riser
4-in installed total	4-in heavy-duty uPVC drainage SN8	303.7340	336 m = 56 pipes x 6 m	10% allowance, rounded to full commercial lengths
Two production return mains	6-in heavy-duty uPVC drainage SN8	78.3822	Included in 6-in total	2 x 39.1911 m
Production tank-room entry allowance	6-in heavy-duty uPVC drainage SN8	2.0000	-	1 m per production tank
6-in installed total	6-in heavy-duty uPVC drainage SN8	80.3822	90 m = 15 pipes x 6 m	10% allowance, rounded to full commercial lengths

654 m
3-in PN16 feed
109 pipes x 6 m

336 m
4-in SN8 return
56 pipes x 6 m

90 m
6-in SN8 return
15 pipes x 6 m

Procurement formula: installed length includes vertical and room-entry allowances. A further 10% cutting and installation allowance is applied, then rounded upward to full 6 m commercial pipe lengths.

Material specification

- **Feed:** uPVC pressure PN16, green identification stripe/tape, solvent-weld joints, suitable for nutrient solution.
- **Buried return:** heavy-duty uPVC drainage SN8 with sealed rubber-ring sockets and blue identification stripe/tape.
- **Exposed table drainage:** uPVC PN10 socket system, mechanically supported on the table chassis.
- Pipe and fittings shall be dimensionally compatible; mixed manufacturer dimensions require an approved transition adapter.

9. Site-Network Fittings and Valves

Item	Unit	Installed	Purchase	Location/function
3 x 3 x 3-in equal pressure tee, PN16	pc	9	10	8 production branches + 1 nursery branch
3-in PN16 long 90-degree elbow	pc	8	9	Approved direction changes and room/riser transitions
3-in full-bore double-union isolation valve	pc	10	11	Feed-group isolation
3-in removable flushing end cap	pc	12	14	Main and branch terminals
3-in pump discharge network interface kit	set	3	4	Flange/union, flexible connector and gauge; check valve belongs to pump package
3-in watertight wall sleeve	pc	3	4	Operating-tank room penetrations
3 x 1-in reducing takeoff to each table	pc	84	93	78 production + 6 nursery tables
1-in full-bore ball valve in table box	pc	84	93	Individual table isolation
1-in union in table box	pc	84	93	Riser removal without cutting
3-in straight repair coupling	pc	0	12	Site emergency spare
6 x 6 x 4-in 45-degree reducing wye	pc	4	5	Two collection branches per production circuit
4 x 4 x 3-in 45-degree reducing wye	pc	84	93	Table downpipe to buried return
6-in long-radius 90-degree elbow	pc	4	5	Production tank-room entries and final changes
4-in long-radius 90-degree elbow	pc	5	6	Three nursery-route elbows plus two entry/coordination elbows
4-in removable cleanout	pc	5	6	Remote ends and changes of direction
6-in cleanout/end cap	pc	2	3	One per production return main
6-in production tank return-entry kit	set	2	2	Sleeve, waterstop, flexible connector, dismantling joint and gate valve
4-in nursery tank return-entry kit	set	1	1	Sleeve, waterstop, flexible connector, dismantling joint and gate valve
4-in repair coupling	pc	0	6	Spare
6-in repair coupling	pc	0	2	Spare
4-in spare rubber sealing ring	pc	0	40	For gasketed return joints
6-in spare rubber sealing ring	pc	0	10	For gasketed return joints

Feed consumables: allow initially for 12 L uPVC pressure cement and 6 L cleaner/primer, subject to the pipe manufacturer's coverage and compatibility requirements.

Valve rule: all feed valves shall be full-bore. Return isolation valves at tanks shall be full-bore gate or butterfly type and shall not be used for throttling.

10. On-Table Drainage Takeoff

Item	Unit	Installed	Purchase	Description
1-in drain branch from NFT drain cap	set	882	971	Cut branch pipe plus cap connector
1-in drain elbow/drop fitting	pc	882	971	Directs flow into collector
3 x 1-in 45-degree branch fitting on collector	pc	882	971	Entry aligned with flow direction
3-in exposed PN10 transverse drain collector	m	168	186	2 m per table
3-in exposed PN10 drain downpipe	m	84	96	1 m allowance per table; 16 pipes x 6 m
3-in removable collector cleanout cap	pc	84	93	Collector terminal
Collector and downpipe support clamps	table set	84	93	Independent support on table chassis
Root/debris trap with removable 1/8-in stainless basket	set	3	3	One before each operating tank

Connection sequence from NFT channel to buried return

1. NFT drain end cap with watertight outlet.
2. 1-in branch and drop elbow into the 3-in collector.
3. Each branch enters through a 45-degree fitting aligned with the collector flow.
4. The 3-in transverse collector falls 1% toward the 3-in downpipe.
5. The downpipe terminates in an accessible box at a 4 x 4 x 3-in 45-degree wye on the buried return.
6. A 3-in removable cleanout remains accessible after table and floor installation.

Mechanical continuity: every cap, elbow, branch and collector shall be physically connected, with no suspended or open pipe ends. Drainage assemblies are clamped independently to the table chassis and remain removable for maintenance.

11. Site Installation Method

Excavation, bedding and backfill

1. Set out centrelines and invert levels and clear conflicts with foundations and electrical routes before excavation.
2. Provide 100 mm compacted clean-sand bedding below the pipe.
3. Maintain at least 400 mm cover above pipe crown in non-traffic areas and 600 mm at service-vehicle crossings.
4. Use selected sand around the pipe and to 150 mm above crown, compacted manually in layers without disturbing slope.
5. Install green warning tape over feed pipes and blue warning tape over return pipes, approximately 200 mm above the pipe.

Jointing

Pressure feed
Square cut, deburr, clean and prime, apply compatible cement, insert to depth mark with quarter turn, then observe the manufacturer's curing time for site temperature.

Gravity return
Clean socket and gasket, apply approved lubricant, align and push to insertion mark, then recheck invert and slope after every joint.

Concrete coordination

- Test, photograph and accept buried piping before aisle concrete is poured.
- Protect every riser in a movement sleeve and cap it during concrete works.
- Locate table valve boxes on the table axis, outside the 0.75 m operating aisle.
- Set box lids flush with finished floor and keep them clear of groundcover.

Heating or flame-forming uPVC is prohibited. Any backfall in a gravity return line, however short, is unacceptable.

12. Testing and Handover

Feed-network pressure test

1. Cap table outlets, fill slowly and vent all high points.
2. Hydrostatically test at 1.5 times operating pressure, or the uPVC manufacturer's stated limit, for two hours after temperature stabilisation.
3. No continuing pressure loss, leakage or joint sweating is accepted. Defective joints shall be remade, not externally patched.
4. Flush from remote ends until clean, then disinfect before connecting the NFT channels.

Gravity-return test

1. Survey invert level every 10 m and at every joint and change of direction.
2. Carry out sectional leakage testing before backfill, followed by a flow test from the farthest table.
3. No standing water shall remain in collectors or mains, and no backup shall occur with all channels operating.
4. Demonstrate removal and cleaning of each root-trap basket.

Handover records

- As-built drawing with pipe code, diameter, length, slope, start/end invert and every buried fitting location.
- Dated photographs before backfilling and concrete placement.
- PN16 and SN8 certificates, adhesive and gasket data, pressure and flow-test reports.
- Valve and chamber identification schedule with normal operating position.
- Physical spare-parts inventory and storage location.

Plumbing Contractor	Site Consultant	Hydrofarms Representative